Boundary Extraction Constrained Siamese Network for Remote Sensing Image Change Detection

Jie Lei, *Member, IEEE*, Yijie Gu, Weiying Xie, *Member, IEEE*, Yunsong Li, and Qian Du, *Fellow, IEEE*

***Abstract*— Change detection (CD) is crucial to the understanding of relationships and interactions among multitemporal high-resolution remote sensing (RS) images. However, various inherent attributes of images have different impacts on CD judgment. How to effectively use helpful information to improve the performance of CD is still a challenge. In this article, we present a boundary extraction constrained Siamese network (BESNet) to dig out the efficacy of boundary information. BESNet is a joint learning network in which a novel multiscale boundary extraction (MSBE) module is embedded. In this way, traditional and deep learning techniques are leveraged to learn together to maximize their respective strengths through cooperation. In particular, a new boundary extraction constrained (BEC) loss function combined with a contractive loss function is used to optimize the BESNet. Considering the interaction between various extracted features, a channel-shuffle fusion strategy is developed to exploit their complementary advantages between features. Our experiments show that the proposed BESNet can significantly improve the CD performance and generate more complete and clearer object boundaries. Experiments conducted on two real datasets over different scenes demonstrate its state-of-the-art performance.**

***Index Terms*— Boundary extractor, change detection (CD), channel shuffle, high-resolution remote sensing (RS) image, Siamese network.**

I. Introduction

REMOTE sensing (RS) image change detection (CD) aims to analyze and detect the differences of the same area imaged at different times [1]. With the wide use of high-resolution RS images and understanding of relationships and interactions between human and natural phenomena [2], numerous applications are beneficial from CD, such as analysis of vegetation cover [3], surface water changes [4], urban planning [5], land resource use management [6], disaster damage assessment [7], street view image analysis [8], medical diagnosis [9], and civil infrastructure [10]. These tasks can make use of bitemporal images as input and produce a pixel-wise CD map, also known as binary CD, which is the main content of this research. Great efforts have been dedicated to determining change pixels.

Traditional CD methods can be divided into two categories: pixel-based CD and object-based CD (OBCD). Pixel-based methods, such as change vector analysis (CVA) [11], principal component analysis (PCA) [12], and Kauth–Thomas transformation (KT, also called tasseled cap transformation) [13], are easy to implement. However, these methods do not take the interactive information between adjacent pixels into account, leading to strong dependence on preprocessing approaches [6]. On the contrary, OBCD methods often leverage image segments to carry on local CD [14]. Nevertheless, suitable parameters are difficult to choose to control the output object size in segmentation algorithms, limiting the performance improvement. In addition, the traditional methods also rely on handcrafted features of RS images. Therefore, existing approaches are insufficient in obtaining high-level change information.

In recent years, as one of the most commonly used deep neural networks, convolution neural networks (CNNs) have demonstrated the advantage of automatically extracting complex high-level features to generate robust CD results. Sakurada and Okatani [15] integrated CNN-based feature extraction and superpixel segmentation to realize street image CD. Building on top of the fully convolutional networks [16], Pablo *et al.* proposed a CD method with deconvolutional networks, which relies on stacking contraction–expansion blocks and the fully convolutional networks. However, it takes the concatenated image pairs as the input for the single-branch network, which may not achieve good feature extraction for individual raw images and may even lead to mutual interference. Fortunately, the Siamese network that shared the same weight parameters is successfully used to evaluate the similarity between two samples by mapping input patterns to an implicitly embedded space [17]. It has attracted great attention for the CD tasks [18]–[20].

Although the preceding methods have achieved promising results, there are still problems that hinder the development of deep learning models for CD. First, various inherent attributes of the image (color, texture, and shape) may have differ-

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ent effects on the CD task due to information validity or pseudo-changes of varying degrees. Second, a change object is susceptible to its surrounding environment, such as illumination, resolution, and noise [8]. These external factors increase the detection difficulty in extracting stable and invariant features. The boundary areas of change objects may be very weak since there are similarities between change objects and the surrounding environment. Third, current deep learning-based methods lack interpretability and fail to obtain particularly satisfactory results due to separate treatment through traditional and deep learning approaches. This would lead to a suboptimal solution because the two ways have not been connected closely to maximize their respective strengths through cooperation. It is also an open problem to be addressed.

With these considerations in mind, we propose a boundary extraction constrained Siamese network (BESNet) to establish a unified network based on traditional and deep learning techniques to fully mine the effective features of boundaries to improve the CD performance. To this end, a novel multiscale boundary extraction (MSBE) module is adopted to extract boundary-related features from the Siamese network. In this way, a cooperation strategy is utilized, where a traditional edge operator is embedded into the deep CNNs to participate in training and process different objects simultaneously. The boundary-enhanced features are fed into deep CNNs to integrate other feature information to achieve semantically meaningful boundaries. In addition, a patch-level second-order nonlocal (PSNL) attention layer [21] is introduced into the Siamese network to better balance the relationship between pixels and highlight the content area of interest. Moreover, the channel-shuffle fusion inspired by ShuffleNet [22] is employed to increase the interactivity between different feature maps and maximum performance gain. Furthermore, we exploit a new boundary extraction constrained (BEC) loss that encourages the predicted CD maps to correctly align with the ground truth (GT), which further promotes the fusion layer to leverage information from former parts. Throughout the training, the BEC loss is combined with a contrastive loss [23] to preferably constrain the network end-to-end.

The main contributions of this article are given as follows.

1) We put forward the BESNet to extract boundary information specially for CD. This design utilizes boundary information as an auxiliary condition embedded into the designed Siamese network. In particular, the BEC loss is introduced to learn BESNet in an end-to-end way to obtain accurate CD results that contain sharper predictions around object boundaries.
2) The MSBE module is developed for boundary extraction, which not only takes the traditionally obtained information as prior knowledge but also utilizes deep learning to obtain hierarchical semantic boundary information with other features.
3) For better joint learning, an appropriate fusion strategy based on channel shuffle is used to explore the interaction between different feature maps, avoiding simple addition and splicing operations, to make the CD performance more accurate.

The remainder of this article is organized into four sections. In Section II, traditional CD and deep learning-based CD are briefly described. Section III introduces the necessary information about the BESNet framework. The experimental results and ablation analysis conducted on several publicly available datasets are presented in Section IV. Finally, Section V draws the conclusion.

II. Related Work

A. Traditional Change Detection

Early CD algorithms use pixel points as the basic unit of image processing, which can be mainly divided into three categories: 1) direct comparison; 2) transformation-based; and 3) classification-based CD. Traditional algebraic methods, such as the residual [24], ratio [25], or linear function [26], are directly used to represent changes. The methods based on image transformation map image spectral information into other feature spaces, thereby enhancing spectral differences and reducing the correlation such as CVA. In addition, with the help of existing image classification algorithms, it is possible to classify bitemporal images, followed by performing CD, such as support vector machine (SVM) [27], decision tree (DT) [28], and artificial neural networks (ANNs) [29]. Techniques based on the idea of object-oriented change detection (OBCD) [14] can produce good results. For example, Chen *et al.* [30] proposed a spatial contrast-enhanced image OBCD approach (SICA) to identify changed areas using shape differences, where a proposed image object detection approach (IODA) is leveraged to segment image and explore intrinsic image objects. Xiao *et al.* [31] presented a new solution to OBCD about building, in which a cosegmentation-based method provides separate and associated multitemporal image segmentation. Although the traditional methods are easy to implement, classification errors and low-level features may reduce the overall CD accuracy.

B. Deep Learning-Based CD

With respect to deep learning-based CD, the existing works mainly take advantage of CNNs, especially the Siamese networks. For example, Zhan *et al.* [32] used the weighted contrastive loss to train a Siamese convolutional network along with a k-nearest neighbor method to promote the initial results. Varghese *et al.* [33] proposed a new parallel architecture composed of ResNet encoding blocks and fully connected deconvolution blocks without the tied weights to localize and identify the changes between image pairs. Motivated by semantic segmentation algorithms, many CD methods use an end-to-end form to improve detection performance. Three fully convolutional neural network architectures (FCNNs) [18], namely, fully convolutional early fusion (FC-EF), FC-Siam-conc, and FC-Siam-diff, discussed different input and ship connection patterns. Moreover, U-shaped network (UNet) ++ was introduced in the CD task to decrease the accumulative error from intermediate processing [34]. Because of availability and interpretability, the attention mechanism has become an

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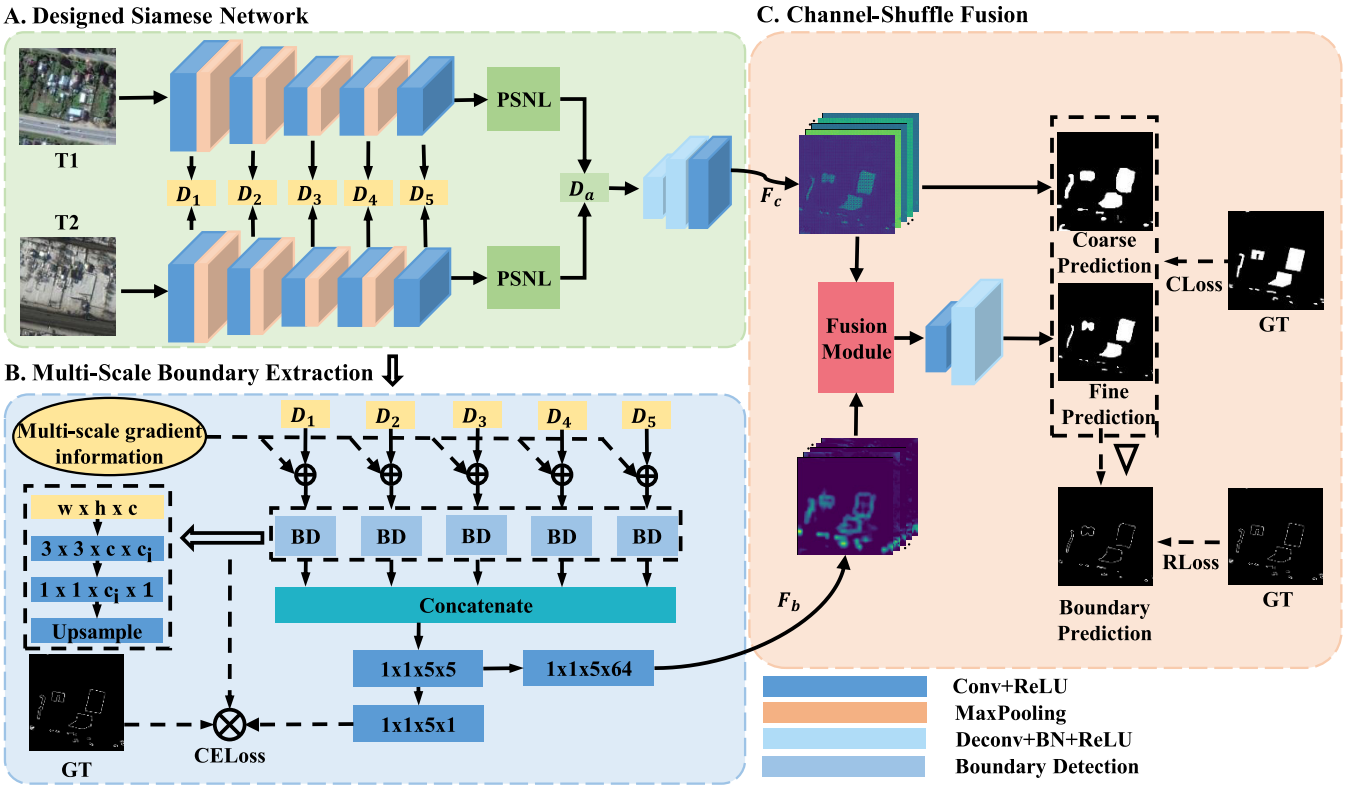


Fig. 1. Schematic of the proposed BESNet-based CD method in RS images. $D_i (i = 1, 2, 3, 4, 5)$ and D_a are obtained difference feature maps. PSNL is an attention module for digging out remote context information. CLoss, CELoss, and RLoss represent the contrastive loss, cross-entropy loss, and regularization loss, respectively. The latter two constitute the total BEC loss function.

active area of the CD. Chen *et al.* [19] utilized spatial–channel attention to obtain better feature representations followed by learning implicit metrics. Chen and Shi [35] designed a CD self-attention mechanism to generate more discriminative features, which employs spatial–temporal relationships in multiscale subregions. A deeply supervised image fusion network (DSIFN) [20] leveraged a series connection between dual attention to effectively overcome the heterogeneity from different features. However, these methods make effort in extracting discriminative features, and most of their architectures draw on the experience of semantic segmentation tasks. Therefore, it is of significance to explore well-directed feature information of RS images to elevate the CD performance.

III. Proposed Method

A. Problem Description

In this article, we mainly study the CD problem for two temporal red-green-blue (RGB) remote sensing (RS) images $I_1, I_2 \in \mathbb{R}^{h \times w \times c}$, where h, w, and c denote the height, width, and the number of image channels, respectively. CD aims to learn a model $f(I_1, I_2)$ that maps the input images into a new map $CM(I_1, I_2) \in \mathbb{R}^{h \times w}$ representing the probability of pixel-level change. Then, using a threshold τ, a binary predict map is obtained, where the values of 1 and 0 represent changed and unchanged pixels, respectively. This process can be summarized as

$$f(I_1, I_2) = \begin{cases} 0, & \text{if } CM(I_1, I_2) < \tau \\ 1, & \text{otherwise.} \end{cases} \tag{1}$$

Based on this basic model, we present our BESNet architecture for CD. As shown in Fig. 1, our network consists of two branches followed by a fusion operation. The designed Siamese network is the first branch that is a basic recognition framework to learn discriminative features. The second branch generates boundary information in the form of deep CNNs in cooperation with traditional operators. In this way, this branch enforces to only process boundary-related information by the carefully designed MSBE module and local supervision. Then, a channel-shuffle fusion is used to fuse the semantic-region features and boundary features to produce a refined detection result, especially around boundaries. Therefore, the change map (CM) $CM(I_1, I_2)$ in our work can be expressed as

$$CM(I_1, I_2) = \Phi(I_1, I_2 | \omega_c, \omega_b, \omega_f) \tag{2}$$

where a nonlinear objective model $\Phi(\cdot)$ is used to transform the original image space into the two classification space, and ω_c, ω_b, and ω_f represent the corresponding parameters of three parts in Fig. 1, which will be explained next.

To verify the efficacy of our idea, two loss functions with contrastive and boundary extraction constraints are incorporated into the learning objective to optimize the proposed network.

B. Designed Siamese Network

This stream, denoted as Φ_c, with parameters ω_c, takes images I_1 and I_2 as input and produces dense pixel features. A baseline extracted network [visual geometry group network (VGG)] and a PSNL attention layer are composed of feature extraction and spatial long-range contextual information.

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According to the size of RS images, the reduction factor of this process has been fine-tuned. Then, a subtraction operation is used to obtain multiscale difference feature maps $D = [D_i, D_a](i = 1, 2, 3, 4, 5)$ between output results of two symmetrical branches, as shown in 3.

Feature maps D_i come from multilayer structures across deep networks, containing features from low to high level, which are further conducive to be fully explored for boundary information. The feature maps D_a have rich semantic meaning in the nature of change through multilevel analysis of data relationships. D_a from the PSNL attention layer are then as the input of a deconvolutional block $\Phi_{\omega_{c2}}$ that is inspired by the SegNet [36] to obtain the output feature maps F_c, as shown in 4. Here, two layers, including the deconvolution (DeConv), batch normalization (BN), and ReLu activation followed by a convolution layer (Conv), are carried out to avoid bias caused by direct upsampling. In particular

$$D = \left\| \Phi_{\omega_{c1}}(I_1) - \Phi_{\omega_{c1}}(I_2) \right\| \tag{3}$$

$$F_c = \Phi_{\omega_{c2}}(D_a | D_a \in D) \tag{4}$$

where $\| \cdot \|$ denotes a similarity measurement strategy that involves a normalization process in the channel dimension to unify the two image feature distributions into the same range for the next subtraction operation.

The optimization process of the parameter ω_c (ω_{c1} and ω_{c2}) can be regarded as a constraint on the difference (d) between two branches in the Siamese network. If a bitemporal pixel pair belongs to changed regions of I_1 and I_2, it is hoped that d is as large as possible. On the contrary, if they are unchanged, then d is small. Therefore, the network has a better feature learning ability to distinguish between change and nonchange pixels. Details of specific constraints will be explained in Section III-E1.

1) Detail of the PSNL Layer: The PSNL attention mechanism performs the second-order statistics on the spatial dimensions of deep semantic feature pairs and highlights the differences between samples at different spatial positions.

Given $F_1 \in \mathbb{R}^{h \times w \times c}$ as the input feature maps for this attention module, F_1 is equally split into k subblocks $F_{1k} \in \mathbb{R}^{h_k \times w_k \times c_k}(k = 1, 2, 3, 4; h_k = h/2; w_k = w/2)$. Two 1×1 convolution layers $\theta(x)$ and $g(x)$ are used to reduce the number of channels c_k to $c_{k/r}$. The value of r follows the configuration of $r = 2$ in [37]. The 3-D outputs are adjusted into 2-D to obtain two tensors B_k and D_k with the size $h_k w_k \times c_{k/r}$. Then, the covariance matrix X_k can be computed using B_k as

$$X_k = B_k \bar{\mathbf{I}} {B_k}^T \tag{5}$$

where $\bar{\mathbf{I}} = 1/c_k/r(\mathbf{I} - 1/c_k/r\mathbf{1})$, which is to follow the practice in [38].

A spatial attention map $X_k \in \mathbb{R}^{h_k w_k \times h_k w_k}$ is generated by performing a global operation on the spatial dimension of B_k to capture the long-range relationship. Then, applying a softmax operation and a matrix multiplication between X_k and D_k to yield

$$U_k = \text{Softmax}(X_k) D_k. \tag{6}$$

-1	0	+1
-2	0	+2
-1	0	+1

(a)

+1	+2	+1
0	0	0
-1	-2	-1

(b)

Fig. 2. (a) Horizontal and (b) vertical kernels of the Sobel operator.

Furthermore, transpose and reshape U_k to the size of $h_k \times w_k \times c_k/r$ and use a 1×1 convolution layer to restore the number of channels to c_k. Finally, similar to the residual architecture, add the output of 1×1 convolution layer $\mathrm{C}_{1\times1}$ and F_{1k} to generate the result of the kth block F_{2k} and combine k subblocks for the final result $F_2 \in \mathbb{R}^{h \times w \times c}$ as

$$F_2 = \mathrm{C}_{1\times1}(U_k) + F_1. \tag{7}$$

Consequently, more robust and distinctive representations can be obtained as our hoped.

C. Multiscale Boundary Extraction

To our knowledge, traditional handcrafted operators use derivatives to calculate pixel differences but are usually not powerful due to their shallow structures. The deep learning networks have rich representations and a strong learning ability of modeling. To extract semantic boundary representations, the proposed MSBE module takes two steps: a multiscale gradient extraction based on a Sobel operator $\Psi(\cdot)$ and a deep learning-based boundary extraction $\Phi_b(\cdot|\omega_b)$.

1) Gradient Extraction Based on a Sobel Operator: Since image edges are high-frequency components in the frequency domain, a simple differential operation may introduce false alarm edges in the extraction results. Therefore, a Gaussian filter is first used to smooth feature maps to reduce the influence of noise. Subsequently, a fixed Sobel operator calculates the gradient information of the previous results in the horizontal and vertical directions.

As shown in Fig. 2, the horizontal and vertical convolution kernels of the Sobel operator are named K_x and K_y, respectively.

Suppose that $F \in \mathbb{R}^{h \times w \times c}$ is the input feature maps and take any channel from it, namely, $F_i \in \mathbb{R}^{h \times w}$. The two convolution kernels perform planar convolution on F_i to obtain horizontal and vertical results on the corresponding single channel

$$g_x = K_x * F \qquad g_y = K_y * F. \tag{8}$$

Then, the final gradient information of feature maps is obtained through a modulus operation

$$g = \sqrt{{g_x}^2 + {g_y}^2}. \tag{9}$$

Here, we perform this Sobel operator $\Psi(\cdot)$ to extract the multiscale boundaries as prior information for the following operation, which makes the MSBE module more compact and learnable.

Let $S_{1i}, S_{2i} \in \mathbb{R}^{h \times w \times c}(i = 1, 2, 3, 4, 5)$ denote the multiscale feature output by the designed Siamese network. First,

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through smoothing processing and the Sobel operator, the corresponding gradients of S_{1i} and S_{2i} can be extracted. Following that, a subtraction is performed to obtain the initial edge difference feature maps G_i, where normalization and absolutization operations can be performed:

$$G_i = \|\Psi(S_{1i}) - \Psi(S_{2i})\|. \tag{10}$$

Then, the multiscale feature difference maps D_i are added to G_i to obtain the preliminary boundary-enhanced difference feature maps $E_i \in \mathbb{R}^{h\times w\times c}(i = 1, 2, 3, 4, 5)$ as

$$E_i = G_i + D_i. \tag{11}$$

The boundary information of each channel in E is enhanced and makes preparations for the following deep learning-based boundary extraction.

2) Deep Learning-Based Boundary Extraction: The deep learning-based boundary extraction focuses only on the associated parts by means of the prior boundary information from the Sobel operator and deep CNNs with powerful loss constraints. The postprocess is composed of the parameter ω_b

$$F_b = \Phi_b(E|\omega_b). \tag{12}$$

As we all know, as the depth of extraction networks increases, image features become more abstract and semantically rich, and shallow features contain more local information, such as edge. A boundary detection (BD) block without deep structures is proposed, which includes two convolution layers $\mathrm{C}_{1\times1}$ and $\mathrm{C}_{3\times3}$ as a key component for boundaries. Let $\Gamma(\cdot)$ denote the function of BD blocks and $i \in 1, 2, 3, 4, 5$ be a multilayer index. Then, the output feature maps b_i of BD blocks can be represented as

$$b_i = \Gamma(E_i) = \mathrm{C}_{1\times1}(\mathrm{C}_{3\times3}(E_i|t)) \tag{13}$$

where t represents the number of output channels in the first convolution layer. The value of t is set to be $64, 32, 16, 8$, and 8 according to the network from low to high to obtain the beneficial boundary information for detection.

Moreover, considering the excellent ability of deep supervision, a constraint is imposed not only for the last output of the boundary extraction branch but also for the multilayer output of BD blocks during training. In order to meet the above requirement, the output channel of $\mathrm{C}_{1\times1}$ is fixed to 1. The detail of the concrete constraint will be explained in Section III-E2.

In order to synthesize the boundary information obtained at different scales, the boundary feature maps F_b are produced by using a concatenation operation on the channel dimension and two 1×1 convolution layers $(\mathrm{C}_1, \mathrm{C}_2)$ to integrate the information differences between the feature maps

$$F_b = \mathrm{C}_2(\mathrm{C}_1([b_1, b_2, b_3, b_4, b_5])) \tag{14}$$

where $[\cdot]$ denotes a concatenate operation. The bilinear interpolation can be used to upsample the feature maps to match the size of the maximum one.

Furthermore, the semantic features F_c and F_b are learned from each other to more accurately distinguish between change and nonchange pixels. Considering that F_c contains more characteristic information, we set the channel number of F_b to half of F_c.

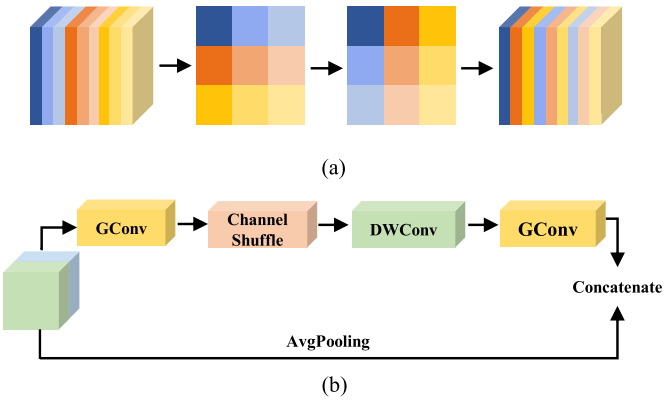


Fig. 3. Diagram of (a) channel-shuffle structure and (b) channel-shuffle fusion block.

Through the above process, the flow of information is facilitated from the Siamese branch to the boundary branch and more dedicated boundary features can be obtained for the following fusion operation.

D. Channel-Shuffle Fusion

When integrating various feature maps with different resolutions and information, most of the existing methods are used directly in combination of Conv+BN+rectified linear unit (ReLU), followed by appropriate postprocessing, such as DeepLabv3+ [39], gated shape CNN (GSCNN) [40], and high-resolution network (HRNet) [41]. Inspired by the ShuffleNet [22], our article uses the concept of channel shuffle to learn more comprehensive feature significance to increase the interactivity. In addition, a depthwise separable convolution [42] and group convolution operation [43] are utilized to further learn spatial and channel correlation information in the fusion process. Then, the corresponding deconvolution and bilinear upsampling operation (Φ_f) will be taken to restore the original size of the input image and obtain the final feature CMs F_f. The first process can be expressed as

$$F_{f1} = \mathrm{CS}([F_c, F_b]|\omega_{f1}) \tag{15}$$

where $\mathrm{CS}(\cdot)$ represents the channel-shuffle fusion block.

The second process utilizes two layers based on 3×3 and 1×1 convolution operation, making use of the parameter ω_{f2} as

$$F_f = \Phi_f(F_{f1}|\omega_{f2}). \tag{16}$$

Besides, we calculate the Euclidean distance [23] between each feature from F_f to gain the CD map F_d as mentioned earlier $(CM(I_1, I_2))$. Finally, a threshold is utilized to obtain the final CD results. Here, the value of the threshold is set to the middle value in each CD result.

1) Detail of the Channel-Shuffle Fusion Block: As shown in Fig. 3(b), the fusion block is divided into two branches: one is used to better complete the information interaction between different feature maps and the other retains the relative information without convolution.

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For the input feature maps F_c and F_b, they are concatenated on the channel dimension and are fed into the fusion block. The upper branch performs a 1×1 group convolution operation, which is characterized by dividing the convolution kernel and the input feature maps into g groups. The group value is set to 2 in this article. Subsequently, divide the channels in each group into several subgroups and provide each group in the next layer with different subgroup information, to achieve channel shuffle. For example, as shown in Fig. 3(a), it can be achieved simply and effectively through reshaping and transposition operations. Assume that there are feature maps with n channels in each group ($n = 3$ and $g = 3$). The output channel dimension is first reshaped to (g, n) and then transpose and flatten back as the input of the next layer, so as to accomplish the anticipated task. A 3×3 depth separation convolution and the second 1×1 group convolution operation are utilized after the channel-shuffle operation to integrate diverse feature information and restore the channel dimension to match the preset value. Besides, the lower branch uses an average pooling operation to lower the spatial dimension, which maintains the translation invariance and robustness of the input feature maps. Finally, a concatenation operation is leveraged to obtain the fused output feature maps.

E. Loss Function

The supervised learning is utilized to optimize the parameters of BESNet with the following two loss functions, from two perspectives of semantic recognition and boundary extraction. In this way, the designed Siamese network and the MSBE are jointly learned together with the channel-shuffle fusion in an end-to-end fashion.

1) Contrastive Loss: To better capture change and nonchange features, it is hoped that the difference between pixel pairs of changed regions is larger, while the difference between unchanged regions is as small as possible. The Euclidean distance d can be utilized to assess the similarity between two image pairs, which can approximate the "semantic similarity" of inputs [23]. Let $v_{1i} \in F_1, v_{2i} \in F_2$ be a pair of pixel vectors in any feature pair F_1 and F_2, and a contrastive loss function is employed to learn the push-and-pull work, which is defined as

$$L_C = \frac{1}{N}\sum_{i=0}^{N-1}\left[-(1-y_i){d_i}^2 - y_i\{\max(0, m-d_i)\}^2\right] \quad (17)$$

$$d_i = \|v_{1i} - v_{2i}\|_2 \quad (18)$$

where N is the number of pixel pairs, $m > 0$ is a margin, and only if the distance between changed pairs is within this margin will it contribute to the loss function. The value of m is set to 2.8 like [19].

2) Boundary Extraction Constrained (BEC) Loss: On the one hand, the cross-entropy loss is utilized to measure the similarity between the prediction of model and GT, widely applied in traditional computer vision tasks. Here, the traditional binary cross-entropy loss is reweighted (w_p and w_n) to enhance boundaries. It can be formulated as

$$L_{\text{bce}} = \frac{1}{N}\sum_{i=0}^{N-1}[-w_p y_i \log(p_i) - w_n(1-y_i)\log(1-p_i)]. \quad (19)$$

where $y_i \in \mathbb{R}^{h\times w}$ denotes GT boundaries, $p_i \in \mathbb{R}^{h\times w}$ denotes the predicted boundary CM of the MSBE module, and w_p and w_n correspond to the weight of loss.

Let n_{pos} and n_{neg} represent the number of boundary and nonboundary samples, respectively. Then, the weight in 19 can be defined as $w_p = n_{\text{neg}}/(n_{\text{pos}} + n_{\text{neg}})$ and $w_n = n_{\text{pos}}/(n_{\text{pos}} + n_{\text{neg}})$. Obviously, n_{neg} is much larger than n_{pos}, so in the actual optimization, more attention is paid to the correctness of pixels at the boundary.

Algorithm 1 CD by the Proposed BESNet

Input: training images $\boldsymbol{I_1}, \boldsymbol{I_2}$, neural network $\boldsymbol{\Phi} = \{\boldsymbol{\Phi_c}, \boldsymbol{\Phi_b}, \boldsymbol{\Phi_f}\}$, a traditional Sobel operator $\boldsymbol{\Psi}$, threshold $\boldsymbol{\tau}$

Output: final detection mask $\boldsymbol{R}$

initialize neural network free parameters ω_c, ω_b, and ω_f randomly;

for each epoch **do**

1: Learn discriminative change features $\boldsymbol{F_c}$ using a designed Siamese network $\boldsymbol{\Phi_c}$ with an attention mechanism from bitemporal images $\boldsymbol{I_1}, \boldsymbol{I_2}$;

2: Extract semantic boundary representations $\boldsymbol{F_b}$ from the multiscale feature outputs by $\boldsymbol{\Phi_c}$ with the help of an MSBE module that consists of a traditional Sobel operator $\boldsymbol{\Psi}$ and a deep learning-based boundary extraction $\boldsymbol{\Phi_b}$;

3: Fuse features $\boldsymbol{F_c}$ and $\boldsymbol{F_b}$ with different significance by a channel-shuffle fusion strategy $\boldsymbol{\Phi_f}$ to generate the CD map $\boldsymbol{F_d}$;

4: Calculate the loss error utilizing a contrastive loss function and a BEC loss function;

5: Optimize the model with Adam;

end

Generate the final detection mask $\boldsymbol{R}$ through the threshold $\boldsymbol{\tau}$ according to (1);

On the other hand, the differences between gradients of $CM(I_1, I_2)$ and GT are computed as the input of a regularization loss to constrain the consistency between them in the boundary space. The gradient means a probability of being a semantic boundary. Similar to [40], this loss can be expressed as

$$L_{\text{reg}} = \frac{1}{2}\sum_{y>thr}\left|\eta(y) - \widehat{\eta}(y_p)\right| + \frac{1}{2}\sum_{y_p>thr}\left|\eta(y) - \widehat{\eta}(y_p)\right| \quad (20)$$

where y and y_p are boundary GT and prediction. $\eta, \widehat{\eta} \in \mathbb{R}^{h\times w}$ represent a potential computed by GT and corresponding change prediction map, which can be realized by the Sober operator in Section III-C1.

Intuitively, we focus on the border of images to ensure that boundary pixels are penalized when there is a mismatch with GT. Besides, a threshold is used to reduce interference from nonboundary pixels, which is set to 10^{-8}. The total boundary extracted constrained loss function (BEC) can be written as

$$L_{\text{BEC}} = L_{\text{bce}} + L_{\text{reg}}. \quad (21)$$

Thus, the overall loss function can be formed as

$$L = \beta_1 L_C + \beta_2 L_{\text{BEC}} \quad (22)$$

where β_i represents the weight for each loss, which is set to 1 for all.

In summary, model $\Phi(\cdot)$ is optimized by minimizing the loss function. When the multinetworks are jointly learned to

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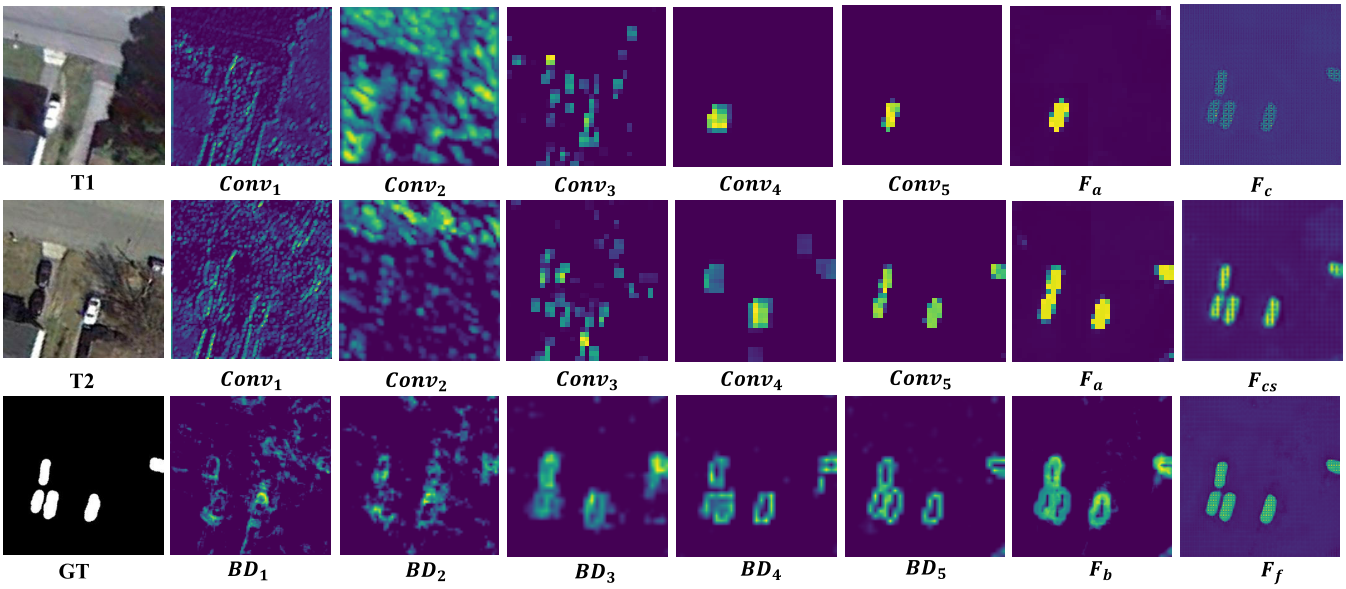


Fig. 4. Visualization of feature maps at different levels of the proposed network. The first column is a research example, including two original images (T1 and T2) and a label map (GT). In Part A, we show the feature maps $\text{Conv}_i(i = 1, 2, 3, 4, 5)$ (backbone) and F_a (attention) of the Siamese network for feature extraction. In Part B, we show the output feature map $BD_i(i = 1, 2, 3, 4, 5)$ of the MSBE module "BD." Part C shows the two input feature maps F_c and F_b of the fusion module F_{cs} after fusion processing and F_f after final deconvolution.

a certain epoch, the parameters ω_c, ω_b, and ω_f of model $\Phi(\cdot)$ are fixed and used for CD. The entire algorithm of BESNet for CD is summarized in Algorithm 1 and we also show the visualization of feature maps at different levels of the proposed network in Fig. 4.

IV. Experiments

A. Data Description

To evaluate the performance of the proposed method, two benchmark datasets are used.

1) CDD: The CD dataset (CDD) [44] is season-vary RS images of the same region with 3–100-cm/px spatial resolution, obtained by Google Earth, which has seven image pairs with the resolution of 4725×2700 pixels for manual GT creation and four image pairs with the resolution of 1900×1000 pixels for adding additional objects manually. It has many diverse object changes with different sizes for binary CD, which ignores the changes caused by seasonal variations. We utilize the dataset patches as in [44], which consists of 16 000 image sets with a size of 256×256 pixels and randomly rotated fragments, and divide into 10 000 training sets, 3000 validation sets, and 3000 testing sets.

2) DSIFN Dataset: This dataset [20] consists of six large coregistered bitemporal image pairs that are collected from six major cities in China provided by Google Earth, including Beijing, Chengdu, Shenzhen, Wuhan, Chongqing, and Xian. The image sizes of these cities are 6236×4652, 4412×3636, 2010×1464, 6963×4555, 6542×5492, and 4392×3140. The dataset ignores diurnal and seasonal changes, while changes between significant land cover types are defined, such as roads and buildings. We select 442 RS images before the augmentation from the official datasets with sizes of 512×512 and divide them according to $8 : 1 : 1$ to generate 358 training sets, 42 validation sets, and 42 testing sets. Then, we use five types of augmentation methods to process the training sets, including flip and rotation.

3) LEVIR-CD Dataset: This dataset [45] contains a total of 31 333 individual image patch pairs with 0.5-m/px spatial resolution obtained by Google Earth. Each image pair has a size of 1024×1024 pixels. These bitemporal images are from 20 different regions in several cities in Texas, USA, taken between 2002 and 2018. It focuses on building-related changes and covers various types of buildings, such as villa residences, tall apartments, small garages, and large warehouses. In practice, we perform nonoverlapping cropping on each image pair to generate four 256×256 patches and filter out the invalid patches to obtain 3167 training sets, 436 validation sets, and 935 testing sets.

B. Evaluation Metrics and Parameter Setting

1) Evaluation Metrics: We assess our proposed method on the testing sets by computing standard performance evaluation metrics, such as precision (P), recall (R), F1 score (F1), overall accuracy (OA), mean intersection over union (mIoU), and IoU for changed samples. Let n_{ij} be the number of pixels, that is, a pixel is predicted as the jth change category but actually belong to the ith change category. The class number is represented by N_c. For these indicators, the larger value means the better performance of algorithms. They are defined as

$$P = \frac{TP}{TP + FP} = \frac{n_{ii}}{n_{ii} + n_{ji}} \tag{23}$$

$$R = \frac{TP}{TP + FN} = \frac{n_{ii}}{n_{ii} + n_{ij}} \tag{24}$$

$$F1 = \frac{2PR}{P + R} \tag{25}$$

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TABLE I
METRIC VALUES FOR ABLATION ANALYSIS OF THE COMPARED BESNET ON THE CDD DATASET

	Methods	P(%)	R(%)	F1(%)	OA(%)	mIoU(%)	IoU(%)
(1)	baseline+A	87.2868	93.7196	90.3889	97.5797	89.8661	82.4632
(2)	baseline+MSBE	92.8738	88.1274	90.4384	97.7371	90.0058	82.5457
(3)	baseline+A+MSBE	95.5100	89.7878	92.5605	98.2473	92.0921	86.1513
(4)	baseline+A+MSBE(wo-t)+CS	95.1814	91.0782	93.0846	98.3567	92.6081	87.0638
(5)	baseline+A+MSBE+CS	**95.2024**	**92.3948**	**93.7776**	**98.5110**	**93.3035**	**88.2842**

A : Attention MSBE : Multi-Scale Boundary Extraction Module wo-t: without traditional methods CS : Channel-Shuffle Fusion Block

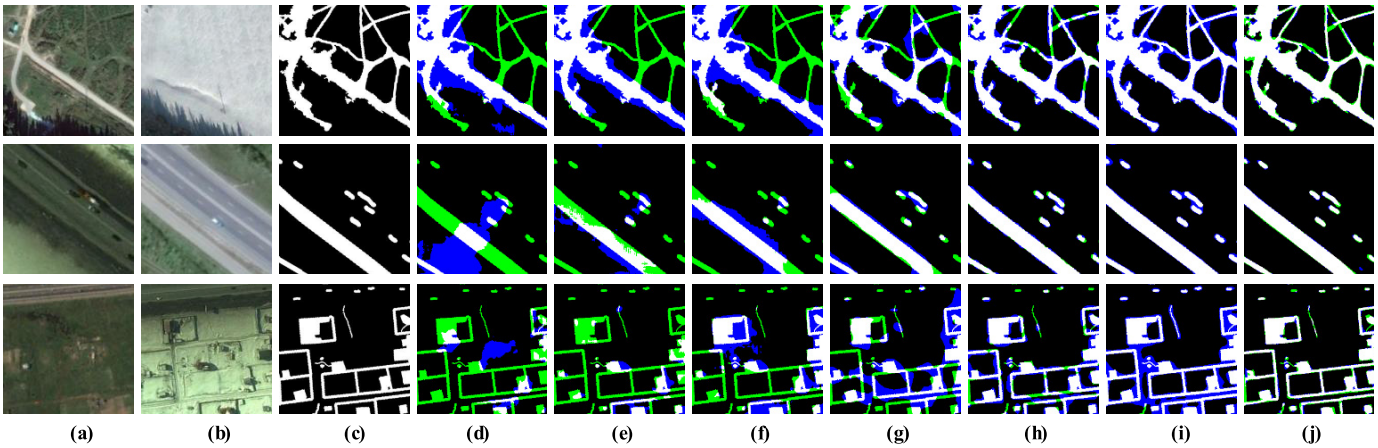


Fig. 5. Visual quality comparison of binary CD on the CDD dataset. (a) T1. (b) T2. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) ChangeNet. (h) dual attentive fully convolutional siamese network (DASNet). (i) IFN. (j) BESNet.

$$OA = \frac{TP}{TP + TN + FP + FN} = \frac{n_{ii}}{n_{ii} + n_{jj} + n_{ji} + n_{ij}} \tag{26}$$

$$\text{mIoU} = \frac{1}{N_c} \sum_{i=0}^{N_c-1} \frac{n_{ii}}{\sum_{j=0}^{N_c-1} n_{ij} + \sum_{j=0}^{N_c-1} n_{ji} - n_{ii}} \tag{27}$$

where P, R, F1, and OA evaluate the changed pixels regions and TP, TN, FP, and FN denote the pixels of true positive, true negative, false positive, and false negative, respectively.

Besides, the receiver operating characteristic (ROC) curve, precision–recall (PR) curve, and areas under them are used. The areas under the ROC curve of [true positive rate (TPR), false positive rate (FPR)] and the PR curve of (P, R) demonstrate the detection accuracy, and the closer this value is to 1, the better the detection capability.

2) Parameter Setting: Experiments of our proposed method are performed on one NVIDIA 2080 Ti graphics processing units (GPUs) with 11-GB memory, which are based on Python 3.6.7, Torch 1.4.0, and compute unified device architecture (CUDA) 10.0. The Adam algorithm is used to train the whole network, and the initial learning rate is set to 10^{-4} with a "poly" learning rate policy, where power $= 0.9$. Besides, the number of cycles for all datasets is selected as 200 epochs for training and the batch size is set to occupy the graphics card memory.

C. Ablation Study

To verify the effectiveness of the attention mechanism, boundary extraction module, and channel-shuffle fusion block, the following comparison scenarios are designed. All the ablation analyses are carried out on the CDD datasets directly and various indicators are tabulated in Table I: 1) the attention mechanism (PSNL) is introduced in the baseline network; 2) the MSBE module is added for extracting boundaries; 3) the combination of attention mechanism and MSBE module is comprehensively evaluated; and 4) and 5) the proposed method with channel-shuffle fusion jointly completes the detection.

By analyzing the following set of scenarios in Table I [group (2, 3)], the attention mechanism achieved a 2.8024% improvement in the mIoU score, and it can be concluded that the attention mechanism is beneficial to the detection. Similarly, the effectiveness of the MSBE module can be obtained from the comparative experiments: group (1, 3). Among them, the system performance is improved by 2.7257%, which indicates that the MSBE module has a positive influence on CD. By adding a channel-shuffle fusion block, the last method in Table I yielded better results, and it was 1.5053% higher than the comparable method 3. Moreover, to verify the effectiveness of the traditional algorithm, we removed it from the method 5. As shown in method 4, the model performance dropped 0.6954%. Numerical results indicate that the utilization of each component during feature extraction and recognition helps to improve the detection results.

D. Experimental Results

We utilize six benchmark methods to evaluate the performance of our algorithm for CD and all these methods are reimplemented according to their articles and open-source codes. For reliable assessments, we test the performance of our network on three different datasets. The visual quality comparison is shown in Figs. 5–7. Tables II–IV list the corresponding metric values. Besides, to better verify the influence of the proposed network on the boundary of detection

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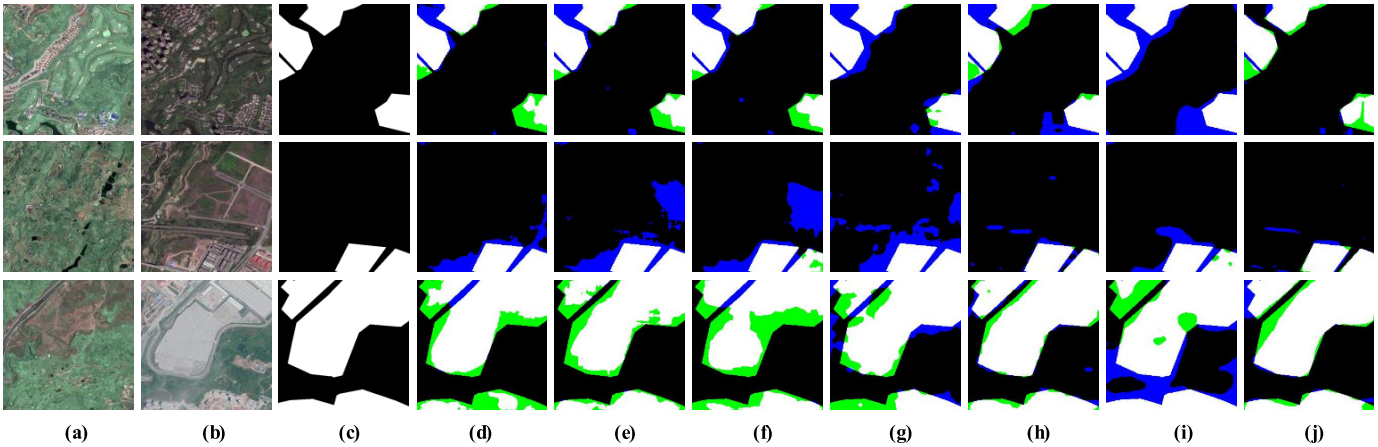


Fig. 6. Visual quality comparison of binary CD on the DSIFN dataset. (a) T1. (b) T2. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) ChangeNet. (h) DASNet. (i) IFN. (j) BESNet.

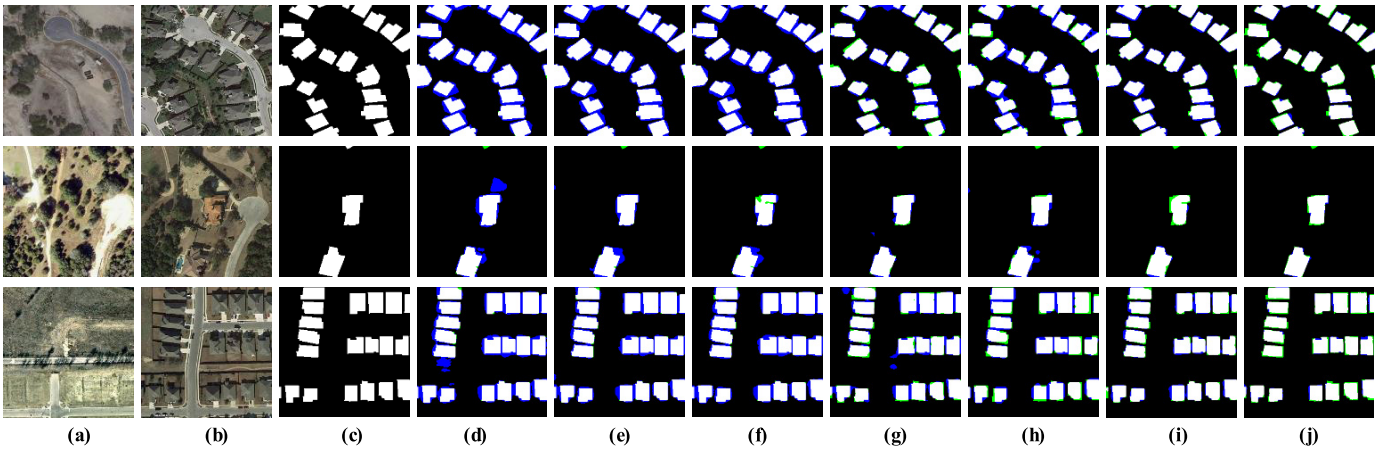


Fig. 7. Visual quality comparison of binary CD on the LEarning, VIsion and Remote sensing laboratory (LEVIR)-CD dataset. (a) T1. (b) T2. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) ChangeNet. (h) DASNet. (i) IFN. (j) BESNet.

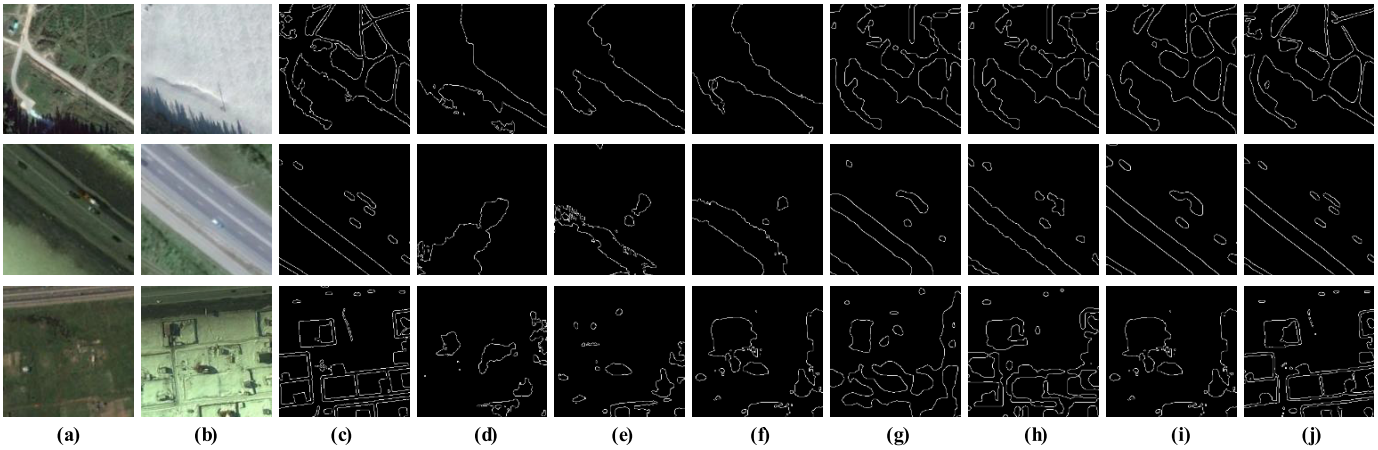


Fig. 8. Boundary visual quality comparison of binary CD on the CDD dataset. (a) T1. (b) T2. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) ChangeNet. (h) DASNet. (i) IFN. (j) BESNet.

results, a traditional boundary extraction operator Canny is utilized to extract boundaries of predictions, which is shown in Figs. 8–10. Moreover, the corresponding ROC curves and PR curves are shown in Fig. 11.

As shown in Figs. 5–7, it represents the visual results of three testing sets, including two original temporal images, GT, six comparison methods, and the proposed model BESNet. To distinguish the pros and cons of various algorithms more clearly and intuitively, the color of white and black in detection results is used to indicate correctly detected samples in the changed and unchanged samples. In addition, from the perspective of change samples as the detection target, green represents missed samples, and blue represents false alarms.

Compared with other benchmark methods, BESNet can produce a structure more similar to the reference image and it can also detect change pixels with different sizes and different

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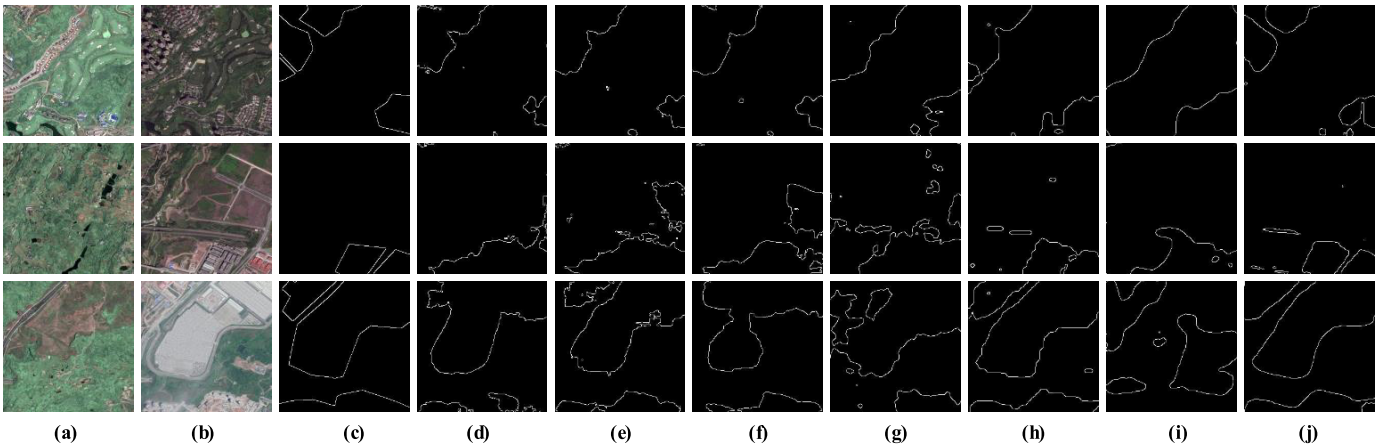


Fig. 9. Boundary visual quality comparison of binary CD on the DSIFN dataset. (a) T1. (b) T2. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) ChangeNet. (h) DASNet. (i) IFN. (j) BESNet.

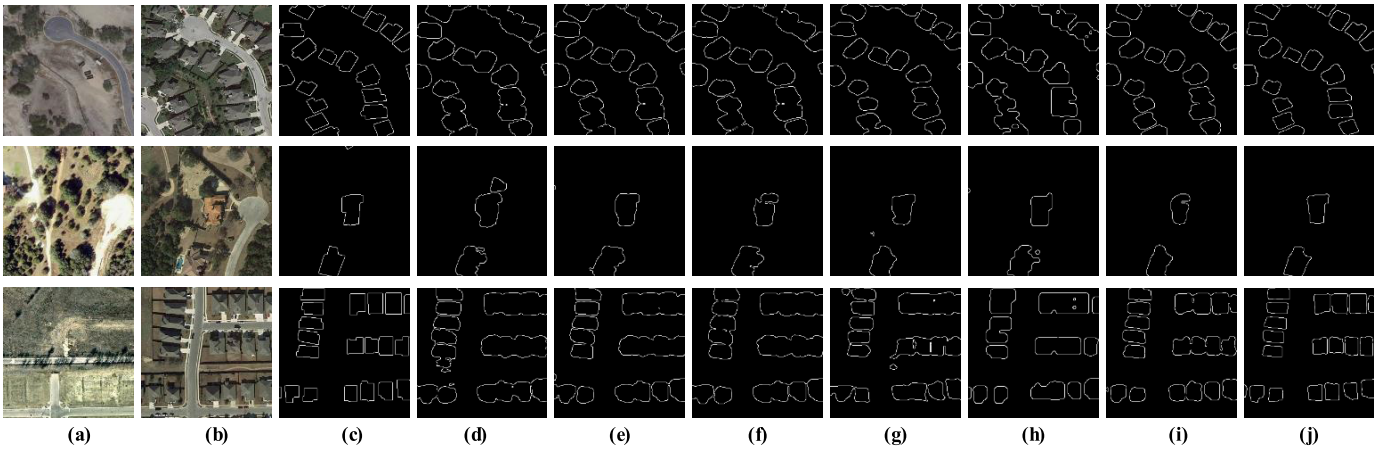


Fig. 10. Boundary visual quality comparison of binary CD on the LEVIR-CD dataset. (a) T1. (b) T2. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) ChangeNet. (h) DASNet. (i) IFN. (j) BESNet.

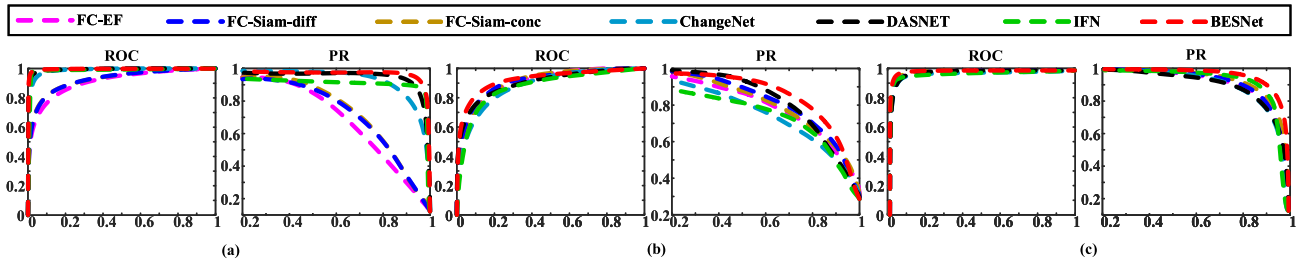


Fig. 11. ROC curves, PR curves, and the corresponding AUC and AP value of different methods on the testing datasets. (a)–(c) Corresponding curves of the CDD, DSIFN, and LEVIR-CD datasets, respectively.

structural information well. The FC-EF, FC-Siam-conc, and FC-Siam-diff methods, compared to other methods, have a poor detection effect on the overall performance, and many changed or nonchanged samples are not correctly classified in most circumstances, resulting in high false alarms and missed detections. ChangeNet and IFN have reduced the problem of false alarms and missed detections on the CDD datasets and the LEVIR-CD dataset to a certain extent, but the thin bar change area detection is not satisfying and they still have obvious detection errors on the second dataset. DASNet performs better in testing datasets but also has more false alarms and rough boundaries of change areas than BESNet. On three datasets, our method can detect the changed area to a greater extent and maintain the integrity of the sample area. According to the metric value of the testing results in Tables II–IV, the mIoU of our method on these datasets can reach 93.3035%, 74.0341%, and 88.8516%, respectively. Moreover, the F1 score can obtain 93.7776%, 77.4679% and 89.0468%, which is also better than other algorithms. Specifically, relative to the mIoU score, it exceeds the second-best method by an average of 2.5094%, 3.1442%, and 0.9529%. In addition, the lower false alarms and missed sample distributions indicate that BESNet can complete the detection task well.

Figs. 8–10 show that the boundaries obtained from the final detection masks generated by BESNet and other compared methods about Figs. 5–7 with the help of the Canny operator. It can be seen that change objects can show more complete and clearer boundaries matching the corresponding labels in

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TABLE II
METRIC VALUES FOR COMPARED METHODS ON THE CDD DATASET

Methods	P(%)	R(%)	F1(%)	OA(%)	mIoU(%)	IoU(%)
FC-EF [18]	59.7668	70.7927	64.8142	90.6662	68.8664	47.9445
FC-Siam-conc [18]	67.8913	72.2905	70.0218	92.4833	72.8160	53.8720
FC-Siam-diff [18]	71.5096	68.5234	69.9847	92.8624	73.0213	53.8280
ChangeNet [33]	86.5951	86.9875	86.7909	96.7846	86.5345	76.6642
DASNet [19]	89.0602	93.7000	91.3212	97.8373	90.7941	84.0286
IFN [20]	84.8105	**97.6475**	90.7774	97.5906	90.1894	83.1123
BESNet	**95.2024**	92.3948	**93.7776**	**98.5110**	**93.3035**	**88.2842**

TABLE III
METRIC VALUES FOR COMPARED METHODS ON THE DSIFN DATASET

Methods	P(%)	R(%)	F1(%)	OA(%)	mIoU(%)	IoU(%)
FC-EF [18]	69.6406	77.7399	73.4677	83.9212	68.6897	58.0624
FC-Siam-conc [18]	66.5188	81.3436	73.1880	82.9336	67.7325	57.7138
FC-Siam-diff [18]	77.3912	72.1315	74.6688	85.9857	70.9574	59.5772
ChangeNet [33]	52.4276	88.0525	65.7229	73.6998	56.8892	48.9457
DASNet [19]	69.5633	79.5437	74.2195	84.1762	69.2580	59.0071
IFN [20]	69.3141	**84.6738**	76.2279	84.9667	70.8899	61.5873
BESNet	**83.5997**	72.1741	**77.4679**	**87.9776**	**74.0341**	**63.2225**

TABLE IV
METRIC VALUES FOR COMPARED METHODS ON THE LEVIR-CD DATASET

Methods	P(%)	R(%)	F1(%)	OA(%)	mIoU(%)	IoU(%)
FC-EF [18]	66.4715	95.0743	78.2407	94.0993	78.8288	64.2585
FC-Siam-conc [18]	70.7570	**95.4875**	81.1626	95.0541	81.3807	68.2971
FC-Siam-diff [18]	70.7163	93.5314	80.5393	94.9564	80.8939	67.4191
ChangeNet [33]	79.2430	87.6235	83.2228	96.0579	83.4485	71.2663
DASNet [19]	76.7460	89.2980	82.5475	95.7867	82.8010	70.2816
IFN [20]	87.8658	88.3582	88.1113	97.3394	87.8987	78.7491
BESNet	**94.4123**	84.2583	**89.0468**	**97.6871**	**88.8516**	**80.2562**

TABLE V
VALUE OF AUC AND AP FOR THE COMPARED METHODS ON THE DIFFERENT DATASETS

Methods	CDD		DSIFN		LEVIR-CD	
	AUC(%)	AP(%)	AUC(%)	AP(%)	AUC(%)	AP(%)
FC-EF [18]	91.0553	72.5835	90.3387	81.2722	98.6711	92.6336
FC-Siam-conc [18]	92.7424	76.2388	90.8954	82.7537	98.9825	94.1798
FC-Siam-diff [18]	92.4834	74.7639	90.9229	83.8252	98.6308	92.9309
ChangeNet [33]	98.8653	93.7935	87.9157	77.1828	97.5195	91.2050
DASNet [19]	99.2637	95.0050	90.0920	84.3108	98.5263	91.4660
IFN [20]	98.7020	91.3826	87.4689	75.8919	97.6467	93.2949
BESNet	**99.5617**	**96.9002**	**92.3642**	**86.6505**	**99.4310**	**96.5272**

our model. This indicates that BESNet can increase the CD performance while maintaining the integrity of the changed target as much as possible.

The ROC and PR curve performance of different CD methods for these three datasets are shown in Fig. 11 and Table V. From the area under the curve (AUC) value and average precision (AP) value, BESNet outperforms the competing methods. In particular, relative to the AUC score of (TPR, FPR), it exceeds the second-best algorithm by an average of 0.298%, 1.4413%, and 0.4485% on three testing datasets. This result is consistent with Figs. 5–7, and Figs. 8–10, which further validates the effectiveness and advantage of the proposed method.

Besides, the average computation time (in milliseconds) on a single image for each comparison method on the CDD dataset is reported in Table VI. The time consumption of these methods is roughly in the same order of magnitude. Considering the corresponding detection performance, the IFN time-consuming ranking and detection performance belong to the high level. The first three methods have shorter test times, but the performance is not satisfactory. Compared with ChangeNet and DASNet methods, the proposed network has a significant performance improvement in the case of a similar time. However, due to the introduction of a boundary extraction branch, our method improves the performance but increases the time consumption. The tradeoff between performance and time consumption has always been one of the important problems in research. This article mainly discusses the significance of boundary for CD, and it is necessary to improve the network structure in the follow-up to reduce time consumption, such as introducing lightweight network design.

TABLE VI
COMPUTING TIME (IN MILLISECONDS) OF THE COMPARED METHOD

Methods	FC-EF	FC-Siam-conc	FC-Siam-diff	ChangeNet	DASNet	IFN	BESNet
Speed(ms)	17	20	18	40	42	26	48

V. CONCLUSION

In this article, we put forward a unified CD method called BESNet. The main objective of our method is to learn a discriminative network that emphasizes the positive influence of the boundary information on CD. Accordingly, the superiority of the BESNet is resulted from several aspects. First, an MSBE module makes full use of the advantages of the traditional and deep learning-based methods to maintain clear boundaries in CD results. Second, in consideration of the interaction between different feature maps, a channel-shuffle fusion strategy is leveraged to autonomously learn more comprehensive features. Furthermore, a new BEC loss is used to improve the discrimination capacity between the boundary and nonboundary pixels. Compared to several deep learning-based CD methods on two publicly available datasets, both qualitative and quantitative performances are improved by the proposed BESNet.

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